

The Stock Trading Simulator

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1 Project Overview

The Stock Trading Simulator is a Java-based application designed to simulate simplified real-world stock trading behavior in a safe and educational environment. The system allows users to register accounts, view simulated stock prices, and execute buy/sell orders with virtual funds. Users can manage their portfolios, monitor their trading history. Administrators can query, modify, delete or export the information of each user. Users can observe the ups and downs of the stock market and the direct comparisons of various stocks. Meanwhile, administrators can also reset the stock market prices. We plan to implement GUIs, Threads, Networking, and Databases in this system.

2 Specification

2.1 login/register

First, run the server and client programs. For the first time, please use the "register" button to initialize a username and password. After that, you can log in using "Login".

2.2 User Menu Guide

The user interface directly displays the current prices of all stocks, along with the user's available funds and stock holdings. The top row of buttons, from left to right, provides the following functions: select stock, view transaction history, buy/sell stocks, proceed to the next day, view info, view stock prices, and view stock price line chart.

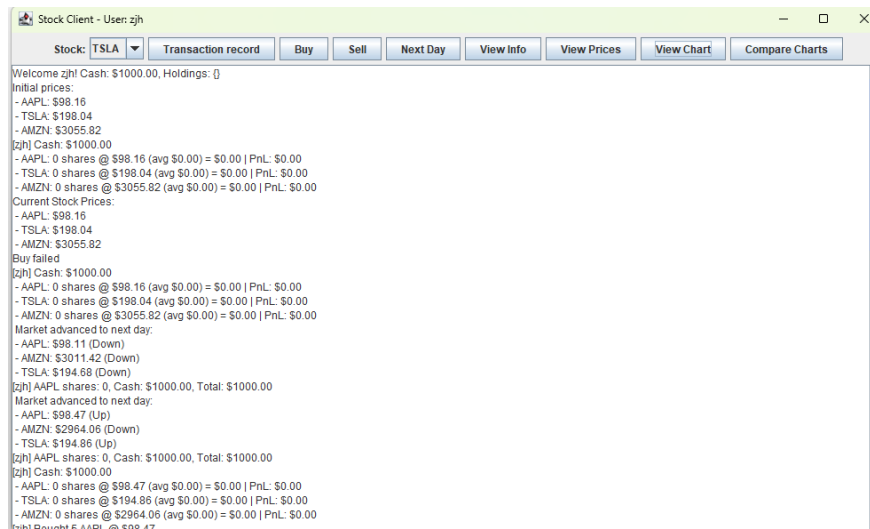


Figure 1: stock client

- **stock selection** At present, users can choose from three available stocks. Stock prices are determined using a hybrid method that combines simulation with actual market change rates.
- **view transaction history:** The transaction record will be displayed after button clicking.
- **buy/sell stocks:** The system allows users to trade a specified number of shares or execute complete buy/sell orders for all available holdings or funds.
- **proceed to the next day:** When proceeding to the next day, the stock prices are updated and users are notified of the price movement, whether it has risen or decreased compared to the previous day.

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Market advanced to next day:
- AAPL: $98.47 (Up)
- AMZN: $2964.06 (Down)
- TSLA: $194.86 (Up)
[zih] AAPL shares: 0. Cash: $1000.00. Total: $1000.00

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Figure 2: next day

- **view info:** Upon clicking the "View Info" button, users can view their current balance, detailed holdings for each stock, the average buying price, and the potential proceeds from selling all held shares at the current market price.

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[zjh] Cash: $326.48
- AAPL: 5 shares @ $97.27 (avg $97.74) = $486.35 | PnL: $-2.35
- TSLA: 1 shares @ $188.92 (avg $186.17) = $188.92 | PnL: $2.75
- AMZN: 0 shares @ $2962.17 (avg $0.00) = $0.00 | PnL: $0.00

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Figure 3: view info

- **view price:** Show the current stock price.
- **view chart and compare chars:** By clicking the "View Chart" button, users can view a line graph illustrating the price trend of the currently selected stock. The "Compare Chart" feature enables users to compare the performance of all available stocks, with the option to use normalized prices for a clearer comparison. All charts can be exported in PNG format for reference or reporting purposes.

2.3 Server Menu Guide

The server interface shows real-time updates of stock price fluctuations and logs every user action performed on the client side. Two buttons at the bottom provide functionality for managing user accounts and resetting stock prices.



Figure 4: view chart



Figure 5: compare charts

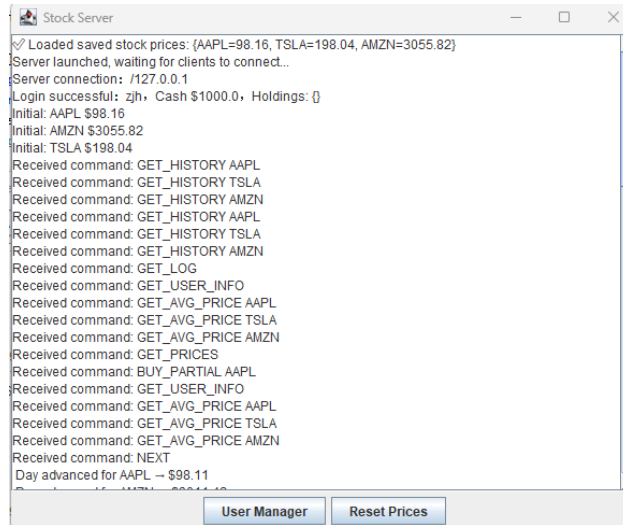


Figure 6: stock server

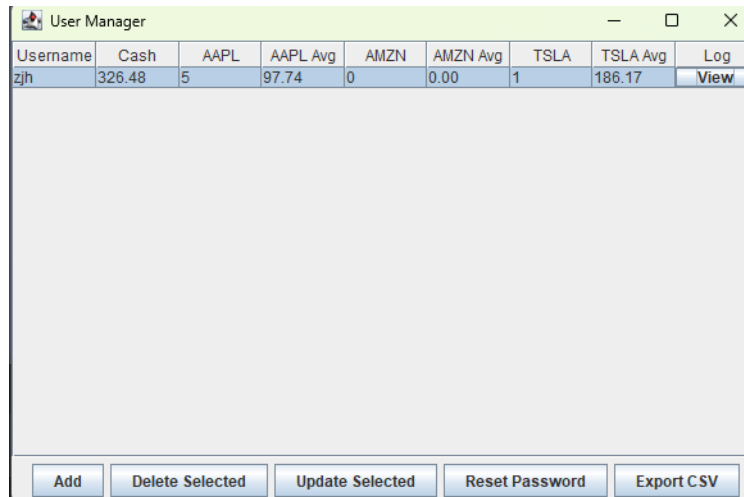


Figure 7: user manager

- **user manager:** Clicking the "User Manager" button opens a new window displaying all current users along with their available funds and stock holdings. Clicking "View" allows you to view the selected user's transaction history. The "Add" button is used to add a new user. After selecting a user, you can delete them or modify specific fields and confirm the changes by clicking "Update Selected." The "Reset Password" button resets the password of the selected user. Clicking "Export CSV" exports the user database as a CSV file.
- **reset prices:** Clicking this button resets all stock prices to their initial values.